Comprehensive Unit-I Basic Electronics Notes: P-N Junction Diode, Zener, Rectifiers, Wave-Shaping, Multipliers, and Varactors

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User prompt: Unit-I: P-N Junction Diode, Depletion layer, Barrier potential, forward and reverse bias, Knee voltage, V-I Characteristics and its Equivalent Models, Avalanche and Zener Break Down, Diode Applications as Half Wave, Full Wave & Bridge Rectifier and their comparative analysis, Clippers, Clampers, Voltage Multiplier Circuit, Zener Diode and its Applications as a voltage regulator, Varactor diode. Give me proper notes extended notes Without leaving topics Give It to me in pdf form

Response: Engineering Electronics: Unit-I Complete Study Notes

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1. P-N Junction Diode Fundamentals

A P-N junction is formed by joining p-type and n-type semiconductor materials within a single continuous crystal lattice.

```
       P-Region                  Depletion Layer                 N-Region
+-----------------------+-----------------------------+-----------------------+
|  Acceptor Ions (-)    |   (-)       |       (+)     |   Donor Ions (+)      |
|  Majority: Holes (o)  |  Fixed      |      Fixed    |  Majority: Electrons(e)|
|  Minority: Electrons  |  Acceptors  |      Donors   |  Minority: Holes      |
+-----------------------+-----------------------------+-----------------------+
                        <---------- W --------------->
                               Electric Field (E)
                                  <---------
```

Formation of Depletion Layer & Barrier Potential

1. **Diffusion Current:** Due to the steep concentration gradient across the junction, holes diffuse from the p-side to the n-side, and free electrons diffuse from the n-side to the p-side.
2. **Recombination and Uncompensated Ions:** When electrons and holes cross the junction, they recombine near the interface. This leaves behind uncovered, immobile bound ions:
   - Negative acceptor ions (N_A^-) on the p-side.
   - Positive donor ions (N_D^+) on the n-side.
3. **Depletion Layer Formation:** A region devoid of free/mobile charge carriers is established near the metallurgical junction. This space-charge region consists strictly of immobile ions.
4. **Internal Electric Field (E_{in}):** The immobile positive and negative ions create an internal electric field directed from the n-region to the p-region.
5. **Drift Current and Equilibrium:** The built-in electric field opposes further majority carrier diffusion while sweeping minority carriers across the junction (drift current). Equilibrium is reached when:

$$I_{\text{diffusion}} = I_{\text{drift}}$$

6. **Barrier Potential (V_0 or V_{bi}):** The electrostatic potential difference developed across the depletion layer at thermal equilibrium prevents further net movement of majority carriers.

The built-in potential is expressed mathematically as:

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Where:

- $V_T = \frac{kT}{q}$ is the thermal voltage ($\approx 25.86 \text{ mV} \approx 26 \text{ mV}$ at $T = 300 \text{ K}$).
- $k = 1.38 \times 10^{-23} \text{ J/K}$ (Boltzmann constant).
- $q = 1.6 \times 10^{-19} \text{ C}$ (Electronic charge).
- N_A, N_D are acceptor and donor impurity doping concentrations.
- n_i is the intrinsic carrier concentration.

Nominal Values at Room Temperature (300 K):

- Silicon (Si): $\approx 0.7 \text{ V}$
- Germanium (Ge): $\approx 0.3 \text{ V}$

Temperature Dependence: The barrier potential decreases with an increase in temperature at an approximate rate of:

$$\frac{dV_0}{dT} \approx -2.0 \text{ mV/}^\circ\text{C to } -2.5 \text{ mV/}^\circ\text{C}$$

2. Diode Biasing & V-I Characteristics

Biasing Configurations

- **Zero Bias ($V = 0 \text{ V}$):**
  - Depletion layer remains at its equilibrium width W_0.
  - Built-in barrier potential opposes majority diffusion. Net current $I = 0 \text{ A}$.
- **Forward Bias ($V_A > V_K$):**
  - Positive terminal of the external DC supply is connected to the p-type region; negative terminal is connected to the n-type region.
  - External electric field opposes the internal built-in field ($E_{ext} \text{ opposes } E_{in}$).
  - Depletion width decreases:

$$W = W_0 \quad 1 - \frac{V_F}{V_0}$$

  - The effective barrier potential drops to $(V_0 - V_F)$.
  - Once V_F approaches the cut-in voltage, majority carriers cross the junction in large numbers, resulting in an exponential rise in forward current (I_F).
- **Reverse Bias ($V_A < V_K$):**
  - Positive terminal is connected to the n-type region; negative terminal is connected to the p-type region.
  - External field aids the internal field (E_{ext} acts in the same direction as E_{in}).
  - Depletion width widens:

$$W = W_0 \quad 1 + \frac{V_R}{V_0}$$

  - Effective barrier increases to $(V_0 + V_R)$, blocking majority carrier flow.
  - A minute reverse saturation current (I_0) flows, carried purely by thermally generated minority carriers (electrons in p-region, holes in n-region).

Knee Voltage (Cut-in / Threshold Voltage V_γ)

The forward voltage point below which the diode current is negligible ($< 1\%$ of rated current), and beyond which the current increases sharply and exponentially.

- $V_\gamma(\mathrm{Si}) \approx 0.6\ \mathrm{V} - 0.7\ \mathrm{V}$
- $V_\gamma(\mathrm{Ge}) \approx 0.2\ \mathrm{V} - 0.3\ \mathrm{V}$

Shockley Diode Equation

The static current-voltage relationship of a p-n junction diode is defined by:

$$I = I_0 \left(e^{\frac{V}{\eta V_T}} - 1 \right)$$

Where:

- I = Total diode terminal current.
- I_0 = Reverse saturation current (doubles approximately every $10^\circ\mathrm{C}$ rise in temperature).
- V = Applied terminal voltage across the diode ($V > 0$ for forward bias, $V < 0$ for reverse bias).
- η = Ideality factor ($\eta = 1$ for Ge; $\eta \approx 1$ to 2 for Si depending on current level; $\eta = 1$ for integrated Si at moderate currents).
- V_T = Thermal voltage ($\frac{kT}{q}$).

Dynamic vs. Static Resistance

- **Static (DC) Resistance (R_D):**

$$R_D = \frac{V_D}{I_D}$$

- **Dynamic (AC) Resistance (r_d):** Determined from the inverse slope of the forward V-I characteristic curve at a specific operating (Q) point:

$$r_d = \frac{dV_D}{dI_D} = \frac{\eta V_T}{I_D + I_0} \approx \frac{\eta V_T}{I_D}$$

  At room temperature for $\eta = 1$:

$$r_d \approx \frac{26\ \mathrm{mV}}{I_D}$$

3. Diode Equivalent Circuit Models

To analyze diode circuits, the non-linear V-I curve is approximated using standard piecewise linear circuit models.

```
1. Ideal Model                2. Simplified (Offset) Model    3. Piecewise Linear Model
      Anode                         Anode                           Anode
        o                             o                               o
        |                             |                               |
      +---+                         +---+                           +---+
      | / | Ideal Switch            | / | Ideal Switch              | / | Ideal Switch
      +---+                         +---+                           +---+
        |                             |                               |
        |                           +---  -                         +---  -
        |                           | | V_gamma                     | | V_gamma
        |                           +---                            +---
        |                             |                               |
        |                             |                              [ ] r_d
        |                             |                              [ ] (Dynamic Res.)
```

```
     |                          |                             |
     o                          o                             o
  Cathode                    Cathode                       Cathode
```

Comparative Summary of Models

Parameter	Ideal Model	Simplified (Offset) Model	Piecewise Linear Model
Equivalent Components	Ideal Switch	Switch + DC Battery (V_γ)	Switch + DC Battery (V_γ) + Resistor (r_d)
Cut-in Voltage (V_γ)	0 V	0.7 V (Si) / 0.3 V (Ge)	0.7 V (Si) / 0.3 V (Ge)
Forward Dynamic Resistance (r_d)	$0 \, \Omega$	$0 \, \Omega$	Included ($r_d = \Delta V / \Delta I$)
Reverse Resistance (r_r)	∞	∞	∞ (or shunt R_r)
Typical Use Cases	High-voltage power circuits ($V_{in} \gg 50 \text{ V}$)	General analog design, clipping, DC supplies	Precision small-signal calculations

4. Reverse Breakdown Mechanisms: Zener vs. Avalanche

When the reverse bias voltage across a p-n junction exceeds a threshold, the reverse current escalates rapidly. This occurs via two fundamentally distinct physical mechanisms.

```
             Current (I)
                  |            Forward Bias
                  |                 /
                  |                /
 Breakdown        |               /
 ------+----------+------------+-----> Voltage (V)
       |          |          V_gamma
       |          |
       |          | Reverse Saturation Region (I_0)
       |
      V_Z
```

Zener Breakdown

- **Doping Concentration:** Very heavily doped on both sides (p^+ and n^+), producing a narrow depletion layer ($W < 10 \text{ nm}$).
- **Mechanism:** Direct quantum mechanical rupture of covalent bonds caused by an intense electric field ($E > 10^6 \text{ V/m}$). Electrons tunnel directly from the valence band of the p-side into the conduction band of the n-side.
- **Breakdown Voltage:** Occurs at relatively low voltages ($V_Z < 4 \text{ V} - 6 \text{ V}$).
- **Temperature Coefficient: Negative** ($\frac{dV_Z}{dT} < 0$). As temperature increases, the crystal lattice energy increases, narrowing the bandgap; covalent bonds break at lower applied voltages.

Avalanche Breakdown

- **Doping Concentration:** Lightly or moderately doped junctions, leading to a wide depletion layer.
- **Mechanism:** Impact Ionization. Thermally generated minority carriers accelerate under the high electric field across the wide depletion layer, acquiring sufficient kinetic energy to knock valence electrons out of covalent bonds through inelastic collisions. These newly liberated carriers accelerate, producing secondary and tertiary collisions (carrier multiplication).
- **Breakdown Voltage:** Occurs at higher voltages ($V_{BR} > 6 \text{ V}$).

- **Temperature Coefficient: Positive** ($\frac{dV_{BR}}{dT} > 0$). Higher temperatures induce increased thermal lattice vibrations, causing carriers to scatter before attaining ionizing kinetic energy; consequently, a higher reverse voltage is required to initiate breakdown.

Comparison Table

Feature	**Zener Breakdown**	**Avalanche Breakdown**
Doping Level	Extremely Heavy	Light to Moderate
Depletion Layer Width	Very Narrow (< 10 nm)	Wide (> 1 μm)
Dominant Mechanism	High electric field tunneling	Impact ionization & collision avalanche
Threshold Voltage	Low (< 5 V)	High (> 6 V)
V-I Knee Sharpness	Very distinct and sharp	Less abrupt initially
Temperature Coefficient	Negative	Positive

5. Rectifier Circuits & Comparative Analysis

A rectifier converts alternating current (AC) into unidirectional pulsating direct current (DC).

1. Half-Wave Rectifier (HWR)

A single diode placed in series with an AC secondary winding and a load resistor R_L.

```
           D1
  o------|>|-----------+
 ~                     |
V_m sin(wt)           [ ] R_L    V_o
 ~                     |
  o--------------------+
```

- **Positive Half-Cycle ($0 \le \omega t \le \pi$):** Diode is forward-biased and conducts. $v_o(t) = V_m \sin(\omega t) - V_\gamma \approx V_m \sin(\omega t)$.
- **Negative Half-Cycle ($\pi \le \omega t \le 2\pi$):** Diode is reverse-biased (open circuit). $v_o(t) = 0$ V.

Governing Derivations

- **Average / DC Current (I_{dc}):**

$$I_{dc} = \frac{1}{2\pi} \int_0^\pi I_m \sin(\omega t) d(\omega t) = \frac{I_m}{2\pi} [-\cos(\omega t)]_0^\pi = \frac{I_m}{\pi}$$

$$V_{dc} = I_{dc} R_L = \frac{V_m}{\pi}$$

- **RMS Current (I_{rms}):**

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^\pi I_m^2 \sin^2(\omega t) d(\omega t)} = \frac{I_m}{2}$$

$$V_{rms} = \frac{V_m}{2}$$

- **Ripple Factor (γ):**

$$\gamma = \frac{I_{ac}}{I_{dc}} = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{I_m/2}{I_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{4} - 1} \approx 1.21$$

- **Rectification Efficiency (η):**

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)} \approx \frac{(I_m/\pi)^2 R_L}{(I_m/2)^2 R_L} = \frac{4}{\pi^2} \approx 40.6\%$$

- **Peak Inverse Voltage (PIV):** The maximum reverse voltage across the diode occurs at $\omega t = \frac{3\pi}{2}$:

$$\mathrm{PIV} = V_m$$

2. Full-Wave Center-Tapped Rectifier (FWR-CT)

Utilizes a center-tapped secondary winding ($V_{s1} = V_m \sin \omega t$, $V_{s2} = -V_m \sin \omega t$) and two diodes.

```
           D1
  o------|>|-----------+
 ~                     |
Center-Tap (GND)       +-------+
  o--------------------+       |
 ~                     |      [ ] R_L    V_o
  o------|>|-----------+       |
           D2                  +
```

- **Positive Half-Cycle ($0 \leq \omega t \leq \pi$):** D_1 is ON, D_2 is OFF. Current flows through R_L from top to bottom.
- **Negative Half-Cycle ($\pi \leq \omega t \leq 2\pi$):** D_2 is ON, D_1 is OFF. Current flows through R_L in the **same direction**.

Governing Derivations

- **Average / DC Current:**

$$I_{dc} = \frac{1}{\pi} \int_0^{\pi} I_m \sin(\omega t) d(\omega t) = \frac{2I_m}{\pi} \implies V_{dc} = \frac{2V_m}{\pi}$$

- **RMS Current:**

$$I_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2(\omega t) d(\omega t)} = \frac{I_m}{\sqrt{2}} \implies V_{rms} = \frac{V_m}{\sqrt{2}}$$

- **Ripple Factor (γ):**

$$\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{I_m/\sqrt{2}}{2I_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} \approx 0.482$$

- **Rectification Efficiency (η):**

$$\eta = \frac{(2I_m/\pi)^2 R_L}{(I_m/\sqrt{2})^2 R_L} = \frac{8}{\pi^2} \approx 81.2\%$$

- **Peak Inverse Voltage (PIV):** When D_1 conducts, anode of D_2 is at $-V_m$ while cathode is at $+V_m$:

$$\mathrm{PIV} = 2V_m$$

3. Full-Wave Bridge Rectifier

Uses four diodes arranged in a closed ring bridge topology, eliminating the need for a center-tapped transformer.

```
            D1            D2
       +----|>|----+--+----|<|----+
       |           |  |           |
AC ~  -+           |  |           +-  ~ AC
       |           |  |           |
       +----|<|----+  +----|>|----+
            D4            D3
```

```
          |              |
          +----[ R_L ]---+
```

- **Positive Half-Cycle ($0 \le \omega t \le \pi$):** Diodes D_1 and D_3 conduct in series. Load current flows through R_L.
- **Negative Half-Cycle ($\pi \le \omega t \le 2\pi$):** Diodes D_2 and D_4 conduct in series. Current enters R_L in the identical direction.
- **PIV:** Across non-conducting diodes (e.g., D_2, D_4 during positive half-cycle):

$$\text{PIV} = V_m$$

Comprehensive Comparative Analysis of Rectifiers

Performance Metric	Half-Wave Rectifier	Center-Tapped Full-Wave	Bridge Full-Wave
Number of Diodes	1	2	4
Center-Tapped Transformer	No	Compulsory	Not required
DC Output Voltage (V_{dc})	$\frac{V_m}{\pi} \approx 0.318V_m$	$\frac{2V_m}{\pi} \approx 0.636V_m$	$\frac{2V_m}{\pi} \approx 0.636V_m$ (minus $2V_\gamma$)
DC Output Current (I_{dc})	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
RMS Output Current (I_{rms})	$\frac{I_m}{2}$	$\frac{I_m}{2}$	$\frac{I_m}{2}$
Rectification Efficiency (η)	40.6%	81.2%	81.2%
Ripple Factor (γ)	1.21	0.482	0.482
Output Ripple Frequency	f_{in}	$2f_{in}$	$2f_{in}$
Peak Inverse Voltage (PIV)	V_m	$2V_m$	V_m
Form Factor ($K_f = I_{rms}/I_{dc}$)	1.57	1.11	1.11
Peak Factor ($K_p = I_m/I_{rms}$)	2.0	$2 \approx 1.414$	$2 \approx 1.414$
TUF (Transformer Utilization Factor)	0.287	0.693	0.812

6. Wave-Shaping Circuits: Clippers & Clampers

Clipper Circuits (Limiters)

Clippers remove portions of an applied signal above or below pre-set reference voltage levels without distorting the remaining segment of the waveform.

1. Unbiased Shunt Clippers

- **Positive Shunt Clipper:** Diode points upward in parallel with load.
  - When $v_{in} > 0$: Diode turns ON, acting as a short circuit $\implies v_o = 0\text{ V}$.
  - When $v_{in} < 0$: Diode turns OFF, acting as an open circuit $\implies v_o = v_{in}$.
  - Result: Positive half-cycle is clipped off.
- **Negative Shunt Clipper:** Diode cathode connected to input.
  - When $v_{in} > 0$: Diode is OFF $\implies v_o = v_{in}$.
  - When $v_{in} < 0$: Diode is ON $\implies v_o = 0\text{ V}$.

- Result: Negative half-cycle is clipped off.

```
Positive Shunt Clipper
     R
 o--[   ]---------o Output
            |
           --- Diode
           / \ (Anode up)
          -----
            |
 o----------+-----o
```

```
Negative Shunt Clipper
     R
o--[   ]---------o Output
        |
      ----- Diode
      \   / (Cathode up)
       ---
        |
o---------+-----o
```

2. Biased Shunt Clippers

By adding a DC reference battery V_R in series with the shunt diode, clipping occurs at specified non-zero thresholds:

- **Positive Bias with Anode Up:** Clips all voltages above $+V_R + V_\gamma$.
- **Negative Bias with Cathode Up:** Clips all voltages below $-V_R - V_\gamma$.

3. Dual (Combinational) Clipper

Combines two anti-parallel diode-battery branches in parallel with the load to restrict the waveform between an upper limit V_{R1} and a lower limit $-V_{R2}$.

```
          R
o-------[   ]-----------+-------------o v_o
                        |
             Branch 1   |   Branch 2
                      +---+   +---+
                      | / |   | \ |
                      +---+   +---+
                        |       |
                      +--- -   - ---+
                     --- V_R1   --- V_R2
                        |       |
o-----------------------+-------+-----o
```

- For $-V_{R2} < v_{in} < V_{R1}$: Both branches remain open; $v_o = v_{in}$.
- For $v_{in} \geq V_{R1}$: Diode 1 conducts; output is clamped flat at $v_o = V_{R1}$.
- For $v_{in} \leq -V_{R2}$: Diode 2 conducts; output is clamped flat at $v_o = -V_{R2}$.

Clamper Circuits (DC Restorers)

A clamper introduces an intentional DC level into an AC input signal without changing the peak-to-peak amplitude (V_{p-p}) of the signal.

Design Rule: The circuit discharge time constant must be significantly larger than the input signal period T:

$$R_L C \gg 10T$$

```
Positive Clamper Circuit
      C
 o---| |----+-------o v_o
            |
          ----- Diode
          \   / (Cathode grounded)
           ---
            |
 o----------+-------o
```

```
Negative Clamper Circuit
      C
o---| |----+-------o v_o
           |
         +---+ Diode
         | / | (Anode grounded)
         +---+
           |
o----------+-------o
```

1. Positive Clamper

- **Operation:** During the first negative peak ($-V_m$), the diode turns ON, acting as a short circuit. The capacitor charges rapidly through the diode's negligible forward resistance to the peak input value:

$$V_C = V_m \text{ (polarity: positive on left, negative on right)}$$

- During subsequent cycles, the diode is reverse-biased and remains OFF. Applying Kirchhoff's Voltage Law (KVL):

$$v_o(t) = v_{in}(t) + V_C = v_{in}(t) + V_m$$

- **Waveform Shift:** Shifts the entire input signal upward by $+V_m$. The minimum output reaches 0 V, and the maximum reaches $+2V_m$. Peak-to-peak voltage is preserved:

$$(V_{p-p})_{out} = 2V_m = (V_{p-p})_{in}$$

2. Negative Clamper

- **Operation:** During the first positive half-cycle ($+V_m$), the diode turns ON. The capacitor charges to peak voltage $V_C = V_m$ (positive on right plate, negative on left plate).

- For subsequent intervals, the diode is OFF. Output is:

$$v_o(t) = v_{in}(t) - V_m$$

- **Waveform Shift:** Shifts the signal downward such that its positive peak is pinned at 0 V, swinging down to $-2V_m$.

3. Biased Clamper

Adding a DC voltage source V_B in series with the diode sets the reference level to V_B instead of 0 V:

- **Positive Clamper with Positive Bias ($+V_B$):** Clamps the lowest negative excursion to $+V_B$.

- **Negative Clamper with Negative Bias ($-V_B$):** Clamps the highest positive excursion to $-V_B$.

7. Voltage Multiplier Circuits

Voltage multipliers are modified clamper-rectifier cascades that generate high DC voltages from low AC sources without requiring expensive high-voltage transformers.

Half-Wave Voltage Doubler

Composed of two diodes and two capacitors.

```
        C1                  D2
  o------||-------+--------|>|---------+-------o (+ 2 V_m)
 ~                |                    |
V_in             --- D1               --- C2
 ~               / \                  ---
  o---------------+--------------------+-------o Ground
```

1. **Negative Half-Cycle:** Diode D_1 is forward-biased, and D_2 is reverse-biased. Capacitor C_1 charges through D_1 to the peak transformer secondary voltage:

$$V_{C1} = V_m$$

2. **Positive Half-Cycle:** Diode D_1 is reverse-biased, and D_2 conducts. The AC source voltage and the voltage on C_1 act in series-aiding polarity:

$$v_{total} = v_{in} + V_{C1} = V_m + V_m = 2V_m$$

   Capacitor C_2 charges through D_2 to this combined peak:

$$V_{C2} = 2V_m$$

3. Across the load (connected across C_2), the continuous steady-state output is:

$$V_{out} = 2V_m$$

Full-Wave Voltage Doubler

Uses two independent capacitor-diode branches connected across the AC line.

- During the positive half-cycle, D_1 conducts, charging C_1 to V_m.
- During the negative half-cycle, D_2 conducts, charging C_2 to V_m.
- Since C_1 and C_2 are configured in series across the output:

$$V_{out} = V_{C1} + V_{C2} = V_m + V_m = 2V_m$$

- Ripple frequency is $2f_{in}$, requiring smaller smoothing filter components than the half-wave topology.

Voltage Tripler and Quadrupler (Cockcroft-Walton Ladder)

By cascading additional diode-capacitor pairs ($C_3 - D_3, C_4 - D_4$):

- C_1 charges to V_m.
- Subsequent capacitors ($C_2, C_3, C_4, \ldots, C_n$) each charge to $2V_m$.
- **Voltage Tripler:** Measured from ground across $C_1 + C_3 = V_m + 2V_m = 3V_m$.
- **Voltage Quadrupler:** Measured across series capacitors $C_2 + C_4 = 2V_m + 2V_m = 4V_m$.

8. Zener Diode as a Voltage Regulator

A Zener diode is designed to operate continuously in its reverse breakdown region, where terminal voltage (V_Z) remains nearly invariant despite large fluctuations in diode reverse current (I_Z).

```
           Series Resistor (R_S)
    o----------[   ]------------+------------+-------o +
 +               --> I_S         |            |
                                 | I_Z        | I_L
Unregulated                     ---          [ ]
DC Input (V_in)                / \ Zener     [ ] Load Resistor (R_L)
                              ----- Diode     |
 -                               |            |
    o----------------------------+------------+-------o -
                                                V_out = V_Z
```

Governing Equations

Applying KCL and KVL to the regulator node:

$$I_S = I_Z + I_L$$

$$V_{out} = V_Z$$

$$I_L = \frac{V_Z}{R_L}$$

$$I_S = \frac{V_{in} - V_Z}{R_S}$$

1. Line Regulation (Varying V_{in}, Constant R_L)

When load resistance R_L is fixed, the load current $I_L = \frac{V_Z}{R_L}$ is constant.

- If V_{in} increases: I_S increases ($\Delta I_S = \frac{\Delta V_{in}}{R_S}$). The excess current is diverted entirely through the Zener diode ($\Delta I_Z = \Delta I_S$). The voltage drop across R_S rises by ΔV_{in}, leaving $V_{out} = V_Z$ constant.
- If V_{in} decreases: I_S decreases; I_Z drops to absorb the change while maintaining I_L and V_{out} constant, provided $I_Z \geq I_{ZK}$ (Zener knee current).

2. Load Regulation (Constant V_{in}, Varying R_L)

When V_{in} is held constant, the total supply current $I_S = \frac{V_{in} - V_Z}{R_S}$ remains fixed.

- If R_L decreases: I_L increases. The Zener diode sheds an equal amount of current ($\Delta I_Z = -\Delta I_L$) such that their sum matches I_S.
- If R_L increases: I_L decreases. The Zener diode draws the excess current to keep I_S and V_Z stable.

Design Criteria: Calculating Limits of Series Resistance R_S

To prevent damage from overheating and avoid loss of regulation:

$$I_{ZK} \leq I_Z \leq I_{ZM}$$

Where $I_{ZM} = \frac{P_{Z(\max)}}{V_Z}$ is the maximum permissible continuous Zener current.

1. **Maximum Allowable Series Resistance ($R_{S(\max)}$):** Occurs at minimum input voltage ($V_{in(\min)}$) and maximum load current ($I_{L(\max)}$):

$$R_{S(\max)} = \frac{V_{in(\min)} - V_Z}{I_{ZK} + I_{L(\max)}}$$

2. **Minimum Allowable Series Resistance ($R_{S(\min)}$):** Occurs at maximum input voltage ($V_{in(\max)}$) and minimum load current ($I_{L(\min)}$ or no load $I_L = 0$):

$$R_{S(\min)} = \frac{V_{in(\max)} - V_Z}{I_{ZM} + I_{L(\min)}}$$

9. Varactor Diode

A varactor diode (also called a varicap, epicap, or tuning diode) is a specialized p-n junction diode operated exclusively under reverse bias to exploit its voltage-variable junction capacitance.

```
      Symbol                     Equivalent Circuit
      Anode                            R_s        L_p
   o-----|>|-----o               o----[    ]----UUUU----+
         | |                                            |
      Cathode                                        +-----+
                                                C_T|  |  | R_p
                                                     +-----+
                                                        |
                                 o----------------------+
```

Physical Operating Principle

Under reverse bias, a p-n junction acts like a parallel-plate capacitor:

- The conductive p and n regions act as capacitor plates.
- The depleted, non-conductive space-charge layer serves as the dielectric medium (ϵ).
- As reverse bias voltage V_R increases, the depletion layer width W widens.
- Since capacitance is inversely proportional to plate separation distance:

$$C_T = \frac{\epsilon A}{W}$$

Increasing V_R widens W, which **decreases** the transition capacitance C_T.

Mathematical Formulation

The voltage-dependent transition capacitance is modeled by:

$$C_T = \frac{C_0}{\left(1 + \frac{V_R}{V_0}\right)^n}$$

Where:

- C_0 = Zero-bias junction capacitance ($V_R = 0 \text{ V}$).
- V_R = Magnitude of applied reverse bias voltage.
- V_0 = Built-in barrier potential ($\approx 0.7 \text{ V}$ for Si).
- n = Doping profile grading coefficient:
  - $n = \frac{1}{2}$ for abrupt (step) junctions.
  - $n = \frac{1}{3}$ for linearly graded junctions.
  - $n = 1$ to 2 for hyper-abrupt junctions (engineered for wide linear frequency-tuning ranges).

Key Equivalent Circuit Parameters

- C_T: Variable transition junction capacitance (typically 2 pF to 100 pF).
- R_s: Series bulk resistance of semiconductor regions (typically $< 1\ \Omega$).
- R_p: Reverse-bias leakage resistance (typically hundreds of $\text{M}\Omega$, usually neglected).
- L_p: Internal package lead inductance.
- **Quality Factor (Q):** Figure of merit defining operational purity at radio frequencies:

$$Q = \frac{1}{\omega C_T R_s} = \frac{1}{2\pi f C_T R_s}$$

Primary Engineering Applications

1. **Voltage-Controlled Oscillators (VCOs):** Adjusts oscillator tank circuit resonant frequency $f_r = \frac{1}{2\pi\sqrt{LC(V)}}$ dynamically via DC control voltages.
2. **RF and Electronic Tuning Systems:** Replaces mechanical variable capacitors in FM receivers, television tuners, and software-defined radio frontend stages.
3. **Phase-Locked Loops (PLLs):** Provides dynamic frequency and phase tracking in modern wireless transceivers.
4. **Parametric Amplification & Frequency Multipliers:** Exploits nonlinear reactive characteristics for low-noise microwave frequency synthesis.

10. Comprehensive Unit-I Quick-Reference Formula Sheet

$$
\begin{aligned}
\text{Built-in Potential:}\quad & V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right), \quad V_T = \frac{kT}{q} \approx 26 \text{ mV at } 300 \text{ K} \\
\text{Shockley Diode Equation:}\quad & I = I_0\left(e^{\frac{V}{\eta V_T}} - 1\right) \\
\text{AC Dynamic Resistance:}\quad & r_d = \frac{\eta V_T}{I_D} \\
\text{HWR Key Ratios:}\quad & V_{dc} = \frac{V_m}{\pi}, \quad V_{rms} = \frac{V_m}{2}, \quad \gamma = 1.21, \quad \eta = 40.6\%, \quad \text{PIV} = V_m \\
\text{FWR Key Ratios:}\quad & V_{dc} = \frac{2V_m}{\pi}, \quad V_{rms} = \frac{V_m}{2}, \quad \gamma = 0.482, \quad \eta = 81.2\% \\
\text{PIV Rating:}\quad & \text{Center-Tap} = 2V_m, \quad \text{Bridge} = V_m \\
\text{Ripple Factor Equation:}\quad & \gamma = \quad \left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1 \\
\text{Zener Regulation Currents:}\quad & I_S = I_Z + I_L = \frac{V_{in} - V_Z}{R_S} \\
\text{Varactor Capacitance:}\quad & C_T = \frac{C_0}{\left(1 + \frac{V_R}{V_0}\right)^n}, \quad Q = \frac{1}{2\pi f C_T R_s}
\end{aligned}
$$
